

Password Based Door Lock System

using Arduino UNO, 4x4 Keypad & Relay Module

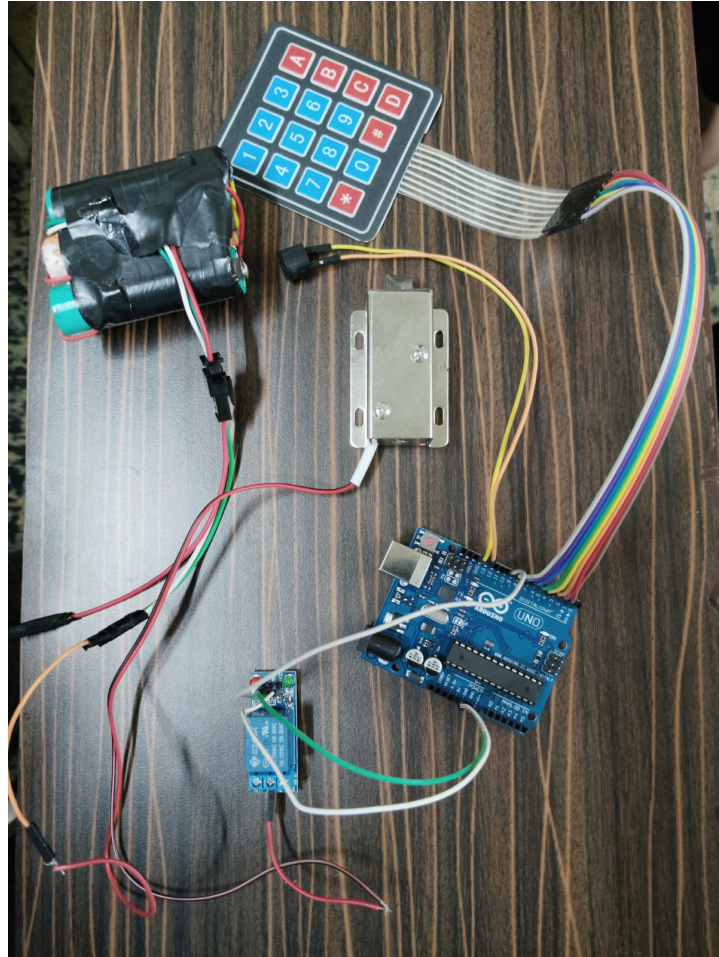


Fig 1: Assembled hardware prototype

Project Report

1. Introduction

This project is a password-protected electronic door lock system built using an Arduino UNO and a 4x4 matrix keypad. The user enters a 4-digit numeric password on the keypad; if it matches the stored password, the Arduino activates a relay that switches power to an electric solenoid door lock, unlocking it. If the entered password is incorrect, a buzzer sounds briefly to alert of the failed attempt.

1.1 Objective

To design a simple, low-cost keypad-based access control system that replaces a traditional mechanical key with a numeric password, using an Arduino microcontroller, a relay-driven electric lock, and basic audible feedback for wrong entries.

2. Components Used

S.No	Component	Quantity	Purpose
1	Arduino UNO	1	Main controller
2	4x4 Matrix Keypad	1	Password entry (0-9, A-D, *, #)
3	1-Channel Relay Module	1	Switches power to the electric lock
4	Electric Door Lock (Solenoid)	1	Physically locks/unlocks the door
5	Buzzer	1	Audible alert on wrong password
6	Battery Pack (2x 3.7V Li-ion)	1	Power supply for the lock/relay circuit
7	Jumper Wires	As required	Connections between modules

3. Circuit Diagram & Connections

The keypad's 4 row pins and 4 column pins connect directly to 8 digital pins on the Arduino. The relay module's signal pin connects to a digital output pin that switches the electric lock's power circuit, while the buzzer is connected to a separate digital output pin for audible feedback.

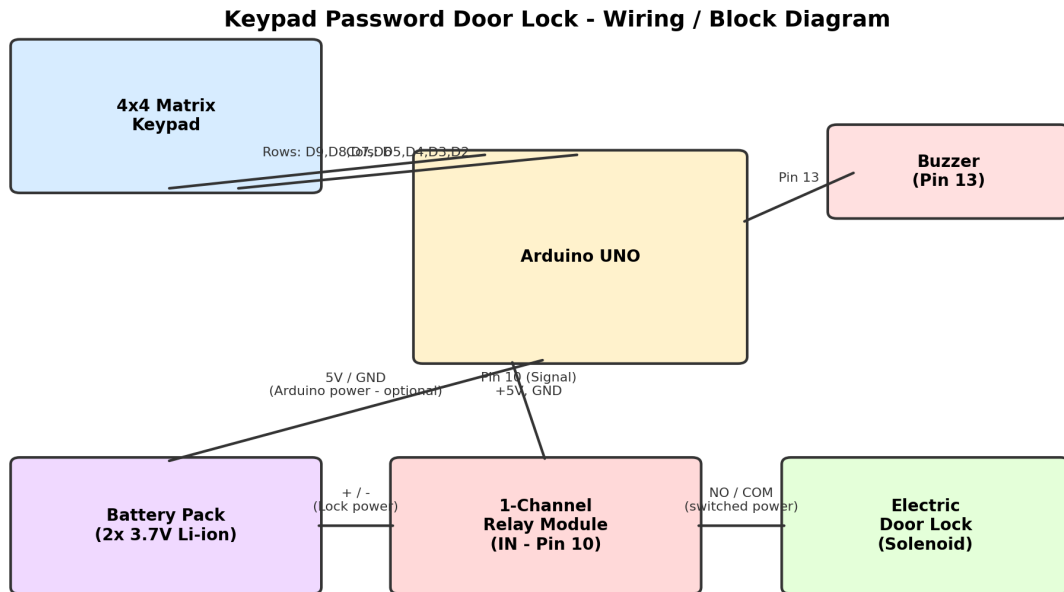


Fig 2: Block wiring diagram of the connections

3.1 Pin Connection Table

Arduino Pin	Connects To	Function
D9	Keypad Row 1	Row scan line
D8	Keypad Row 2	Row scan line
D7	Keypad Row 3	Row scan line
D6	Keypad Row 4	Row scan line
D5	Keypad Col 1	Column scan line
D4	Keypad Col 2	Column scan line
D3	Keypad Col 3	Column scan line
D2	Keypad Col 4	Column scan line
D10	Relay IN (Signal)	Turns relay ON/OFF -> unlocks door
D13	Buzzer +	Sounds alert on wrong password
5V / GND	Relay VCC / GND	Relay module power
-	Battery + / Relay COM-NO / Lock	Switched power path to the solenoid lock

4. Working Principle

On power-up, the Arduino initializes the relay pin and buzzer pin to LOW (off/locked state). The keypad is continuously scanned in the main loop using the Keypad library, which detects which key is pressed by cycling through the row and column pins.

Every digit (0-9) pressed by the user is appended to an input string. Once 4 digits have been entered, the input is compared against the stored 4-digit password ("1234" in this code):

- Correct password: the relay pin (D10) is set HIGH, energizing the relay and switching power to the electric door lock, which unlocks the door.
- Incorrect password: the buzzer pin (D13) is set HIGH for 1 second and then turned off, producing an audible alert for the failed attempt.
- In both cases, the input string is cleared afterward so the next 4-digit attempt can be entered.

5. Source Code (Arduino IDE)

The code below runs on the Arduino UNO. It reads keypad input, checks it against the stored password, and drives the relay (lock) or buzzer accordingly.

```
#include <Keypad.h>

const byte ROWS = 4;
const byte COLS = 4;

char keys[ROWS][COLS] = {
  {'1', '2', '3', 'A'},
  {'4', '5', '6', 'B'},
  {'7', '8', '9', 'C'},
  {'*', '0', '#', 'D'}
};

byte rowPins[ROWS] = {9, 8, 7, 6};
byte colPins[COLS] = {5, 4, 3, 2};

Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS, COLS);

String password = "1234";
String input = "";

void setup() {
  pinMode(10, OUTPUT); // Relay
  pinMode(13, OUTPUT); // Buzzer

  digitalWrite(10, LOW);
  digitalWrite(13, LOW);
}

void loop() {
  char key = keypad.getKey();

  if (key) {
    if (key >= '0' && key <= '9') {
      input += key;
    }

    if (input.length() == 4) {
      if (input == password) {
        digitalWrite(10, HIGH); // Relay ON
      } else {

```

```
digitalWrite(13, HIGH); // Buzzer ON
delay(1000);
digitalWrite(13, LOW);

}

input = ""; // Clear input for next attempt
}
}
}
```

6. Setup & Connection Steps

- 1 Install the 'Keypad' library in the Arduino IDE (Sketch -> Include Library -> Manage Libraries -> search 'Keypad' by Mark Stanley/Alexander Brevig).
- 2 Upload the sketch above to the Arduino UNO.
- 3 Wire the keypad's 8 ribbon-cable pins to Arduino digital pins D9, D8, D7, D6 (rows) and D5, D4, D3, D2 (columns) as per the pin table in Section 3.1.
- 4 Connect the relay module's signal pin to D10, and its VCC/GND to the Arduino's 5V and GND.
- 5 Wire the battery pack, relay's switched output (COM/NO), and the electric door lock in series so the relay controls power to the lock.
- 6 Connect the buzzer's positive terminal to D13 and negative terminal to GND.
- 7 Power the Arduino; enter the 4-digit password (default '1234') on the keypad to unlock the door, or an incorrect code to trigger the buzzer.

7. Conclusion

This project successfully demonstrates a basic password-based door lock system using an Arduino UNO, a 4x4 matrix keypad, a relay module, and an electric solenoid lock. It replaces a conventional key with a simple numeric password, giving controlled access while providing immediate audible feedback for incorrect attempts through the buzzer. The project illustrates key embedded systems concepts such as matrix keypad scanning, relay-based switching of higher-power loads, and simple conditional access control logic.

The system can be further enhanced by adding an LCD display to show input status, an EEPROM-based password change feature, a limited-attempt lockout with timed delay for security, a keypad lock auto-relock timer, or an app/RFID-based secondary authentication method. Overall, the project offers a practical and affordable introduction to microcontroller-based access control and security systems.